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Precise Synthesis of π -Conjugated [3]Catenanes and Solomon Link for Photothermal Responses via a Dual-Tuning Strategy

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Abstract

Mechanically interlocked supramolecular compounds have garnered widespread attention but it remains challenging to investigate topological adjustments in such complicated structures. Here, the authors use a bidentate pyridyl ligand featuring an extended π -conjugated plane for coordination-directed self-assembly to engineer three different kinds of sophisticated complexes, including three Linear [3]catenanes **1**(OTf)₁₂, **2**(OTf)₁₂ and **3**(OTf)₁₂, two Borromean rings **4**(OTf)₁₂ and **5**(OTf)₁₂, and one heterometallic Solomon link **6**(OTf)₈ by modulating the Ru $\cdots$ Ru distances and conjugation planes of the building units or incorporating bridging silver ions. Subtle alterations in the building units can yield a diversity of topological architectures. Differences in the photothermal conversion properties of **1**(OTf)₁₂ - **6**(OTf)₈ are investigated. Various photothermal conversion efficiencies (23% - 81%) are obtained because of the synergistic effect of $\pi\cdots\pi$ stacking interactions and the free radical effect of building units. This finding enables the precise regulation of topology, thereby leading to a remarkable enhancement in photothermal conversion efficiency.

Introduction

Mechanically interlocked supramolecular compounds have garnered widespread attention,

primarily because of their intriguing structures and broad application prospects across various fields.¹⁻⁵ In recent years, an increasing number of interlocked supramolecular compounds, covering rotaxanes,⁶ [2-6]catenanes,⁷⁻¹¹ interlocked cages,¹²⁻¹⁵ ravels^{16,17} or molecular knots,¹⁸⁻²⁰ have been constructed through coordination-driven self-assembly, metal ion templating, or hydrogen bonding interactions, etc, promoting major advances in synthetic chemistry and biological molecule simulations. However, the unique topological advantages and functional differences of the reported topologies have rarely been explored because these structures are often synthesized based on different subcomponents, which may introduce many variable factors. Therefore, a more effective approach, termed the "two-component construction strategy" has been proposed. This strategy involves using an identical subcomponent unit to construct different topological architectures by varying the composition of the second component, thus enabling a comparative investigation of their characteristic differences.

Significant advances in the synthesis of multiple sophisticated interlocking molecules based on the same ligand precursors have been made by Leigh *et al.*,^{21,22} Fujita *et al.*,^{23,24} Stoddart *et al.*,^{25,26} and others.²⁷⁻²⁹ Recently, Jin *et al.* utilized various rigid angular, semi-flexible conjugated, or asymmetric pyridyl ligand precursors in conjunction with rhodium-based building units to synthesize and characterize 8₁₈ knots and Borromean rings, Solomon links and trefoil knots, as well as 4₁ knots and double trefoil knots, demonstrating that two topological species can be obtained by changing the rhodium-based building units based on the same pyridine ligand.³⁰⁻³² Interestingly, Sun *et al.* achieved structural changes between different interlocked cages based on the positive polypyridine ion ligands.³³⁻³⁵ Furthermore, Han *et al.* developed a series of metallacyclic building units that incorporate N-heterocyclic metal carbenes, π -cyclopentadienyl, and π -cycloheptatrienyl motifs, establishing an innovative strategy to form unprecedented organometallic supramolecular structures.³⁶⁻³⁹ While previous work has achieved diverse topologies by varying the linker ligand. In contrast, constructing multiple complex architectures from a single ligand by fine-tuning other intriguing topological species remains rare.

To our knowledge, Linear [3]catenanes,^{40,41} Borromean rings,⁴²⁻⁴⁴ and Solomon links⁴⁵ exhibit

distinct entanglement patterns. The Linear [3]catenane consists of three interlocked rings arranged in a linear sequence. In contrast, the Borromean rings constitute a highly symmetrical structure of three rings, characterized by their collective interlocking without any pairwise interlocking. Conversely, the Solomon link is a complex entity formed from just two macrocycles, each adopting a figure-eight conformation and doubly interwoven with the other. A careful analysis of the three kinds of topologies revealed that the two-component assembly strategy, which is based on the dinuclear building unit and the connecting ligand, is expected to simultaneously achieve the construction of three types of topological compounds. As we know, the synthesis of Linear [3]catenanes and Borromean rings necessitates the use of straight, rigid connecting ligands. Therefore, on the basis of a rigid and linear ligand, controlling the size and conjugation of dinuclear building units to achieve stacking interactions between connecting ligands is expected to lead to the formation of Linear [3]catenane. Meanwhile, increasing the size and conjugation properties of the dinuclear building unit to result in stacking interactions between the ligand and the building unit might generate a Borromean ring structure. However, the preparation of a Solomon link, which consists of double interlocked rings, typically requires a flexible or angular connecting ligand. To construct a Solomon link using a rigid and linear connecting ligand, it might be necessary to employ building units with torsional characteristics. Therefore, simultaneously constructing the construction of the three types of topologies based on a rigid connecting ligand might require the selection of three different construction units (those with weak conjugation, strong conjugation and distortion characteristics).

Half-sandwich organometallic compounds exhibit specific coordination sites, coordination directions and tunable electronic, steric, and catalytic properties, making them ideal components for the construction of complicated structures with specific performance.⁴⁶ Typically, the p-cymene Ru unit can act as a versatile platform for the assembly of functional complexes. Therefore, the three topological structures (Linear [3]catenane, Borromean ring, and Solomon link) are expected to be realized by adjusting p-cymene Ru building units based on a rigid and linear ligand.

In this work, a unique rigid pyridyl ligand precursor, 2,5-di(pyridin-4-yl)thiazolo[5,4-*d*]thiazole

(**L1**) was utilized to synthesize three kinds of different topologies,⁴⁷ covering three Linear [3]catenanes, two Borromean rings, and one heterometallic Solomon link. Note that the thiazolo[5,4-*d*]thiazole unit of **L1** functions as an effective π -conjugated core and an electron-rich moiety. Moreover, the pyridine groups provide two coordination sites and an electron-deficient unit. The building units **E1**(OTf)₂, **E2**(OTf)₂, and **E3**(OTf)₂, which possess slightly shorter distances and weak conjugate effects, induced stacking interactions between the central unit and the pyridine unit of different ligands **L1**, resulting in the generation of three Linear [3]catenanes. The building units **E4**(OTf)₂ and **E5**(OTf)₂ with longer sizes and greater conjugation resulted in stacking interactions between the building unit and the central unit of ligand **L1**, leading to the formation of two Borromean rings. Interestingly, the addition of two bridging silver ions resulted in a twisted building unit **E6**(OTf)₂, which led to the formation of a heterometallic Solomon link. The formation of all the structures was confirmed by Single-crystal X-ray diffraction (SC-XRD) analysis, Nuclear Magnetic Resonance (NMR) spectroscopy, and Electrospray Ionization-Time of Flight Mass (ESI-TOF-MS) analysis. Subsequent photothermal conversion studies were conducted using the obtained compounds, and compound **4**(OTf)₁₂ demonstrated the most significant photothermal response, with temperatures increasing from 25°C to 64.5°C and conversion efficiencies ranging from 77% to 81%. The exceptional photothermal conversion efficiency of **4**(OTf)₁₂ was attributed to the strong $\pi \cdots \pi$ stacking interactions between its building blocks and central unit of ligands, as well as to the free radical effect of the half-sandwich fragment. Furthermore, the results of the EPR tests confirmed that more electrons were excited in compound **4**(OTf)₁₂ before photostimulation than after, with the signal intensity increasing by 11.66 times.

Results

Self-assembly of Linear [3]catenanes **1**(OTf)₁₂, **2**(OTf)₁₂, and **3**(OTf)₁₂.

In general, building units endowed with short, conjugated planes possess minimal spatial steric hindrance and exhibit a relatively weak conjugation effects. Therefore, their capacity to establish robust $\pi \cdots \pi$ stacking interactions with the conjugated centre of the linker ligand precursor **L1** is constrained. Additionally, the donor electron property of thiazolo[5,4-*d*]thiazole centre and the

electron deficiency property of two pyridine units of **L1** enable the formation of the stacking interactions. Thus, the selection of building units with short sizes and weak conjugation effects enables the formation of Linear [3]catenanes, in which a metallamacrocyclic is interlocked with two additional metallamacrocyclics. A mixture of **L1** (0.12 mmol) and **E1**(OTf)₂ (0.12 mmol) in methanol at ambient temperature was stirred for 12 h to yield a dark earthy brown solution. A single crystal of **1**(OTf)₁₂ suitable for X-ray diffraction analysis was obtained by slowly diffusing diethyl ether into a saturated solution of compound **1**(OTf)₁₂ in methanol/diisopropyl ether. **2**(OTf)₁₂ and **3**(OTf)₁₂ were obtained similarly to **1**(OTf)₁₂ by using **E2**(OTf)₂ and **E3**(OTf)₂ instead of **E1**(OTf)₂, and they were isolated as dark green and brown solids in high yields (**1**(OTf)₁₂: 89%, **2**(OTf)₁₂: 90%, and **3**(OTf)₁₂: 91%, Fig. 1).

The structures of compounds **1**(OTf)₁₂, **2**(OTf)₁₂ and **3**(OTf)₁₂ were confirmed by single-crystal X-ray diffraction analysis. All three complexes crystallize in the triclinic $P\bar{1}$ space group (Supplementary Tables 1, 2 and 3). Compounds **1**(OTf)₁₂, **2**(OTf)₁₂, and **3**(OTf)₁₂ were identified as Linear [3]catenanes, which were constructed by intramolecular $\pi\cdots\pi$ stacking interactions and steric effects. In the structure of compound **1**(OTf)₁₂, three mutually interpenetrating, rectangular metallamacrocyclics form a discrete and indivisible ensemble. Each metallamacrocyclic is defined by edge lengths of 16.99 Å and 7.92 Å (Supplementary Figs. 5a, 6a and 7a). Compounds **2**(OTf)₁₂ and **3**(OTf)₁₂ were halogen-substituted derivatives of **1**(OTf)₁₂, where the hydrogen atoms on the phenyl groups of the building blocks were replaced by halogen atoms. Despite modifications to the building blocks, the Ru $\cdots$ Ru distance remained constant, resulting in no alteration to the structural type. Each single molecule interacts with its two adjacent counterparts, thereby establishing a structural connection that does not involve direct chemical bonding (Fig. 2a, b). The molecular structure of compound **1**(OTf)₁₂ revealed that the distances between each ring ranged from 3.49 Å to 3.60 Å. Note that multiple $\pi\cdots\pi$ stacking interactions constituted dodecanuclear supramolecular compound **1**(OTf)₁₂ with side dimensions of 25.52 Å and 28.25 Å (Fig. 2c, d). Similarly, the molecular structure of compound **2**(OTf)₁₂ exhibited distances ranging from 3.44 Å to 3.64 Å between rings and side distances of 24.47 Å and 27.13 Å. Structural analysis revealed that compound **3**(OTf)₁₂ features ring separations spanning from 3.35 Å to 3.68 Å, with overall molecular widths of 23.54 Å and 27.39 Å on the sides (Supplementary Figs. 2, 3, 5b, 5c, 6b, 6c, 7b and 7c). Furthermore, space-filling representations revealed the spatial relationship of each metal ring (Fig. 2e). These high-resolution structural representations delineate the precise

geometric configurations of individual metallacycles and demonstrate their relative orientations, intermolecular distances, and packing patterns in the solid state.

To investigate the behaviour of compounds **1**(OTf)₁₂, **2**(OTf)₁₂, and **3**(OTf)₁₂ in solution, the ¹H NMR, ¹H-¹H COSY NMR, and ¹H DOSY NMR spectra of these compounds were recorded. In the ¹H NMR spectrum of compound **1**(OTf)₁₂ (Supplementary Fig. 15), the signals at 8.44, 8.41, 8.38, 8.03, 7.72 and 7.39 ppm were assigned to the pyridyl protons of ligand **L1**. Furthermore, the resonances at 6.02 and 5.78 ppm were attributed to the benzoquinone protons of the building unit **E1**(OTf)₂. Moreover, the protons of the p-cymene moiety displayed ten characteristic signals at 6.17, 6.05, 5.95, 5.83, 3.00, 2.90, 2.31, 2.20, 1.46, and 1.38 ppm. This multiplicity in the NMR spectrum is attributed to the uniquely entangled topology of these compounds, which creates distinct proton environments within the same chemical moiety. All the resonances were assigned using the ¹H-¹H COSY NMR spectrum of compound **1**(OTf)₁₂ (Supplementary Fig. 16). Interestingly, the presence of a single diffusion signal in the ¹H DOSY NMR spectrum of **1**(OTf)₁₂ ($D_1 = 2.42 \times 10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$) (Supplementary Fig. 17) confirmed the homogeneity of the solution, with the calculated hydrodynamic radii (15.5 Å) closely matching the crystallographic dimensions of the molecule (Supplementary Fig. 50). For compound **2**(OTf)₁₂ (Supplementary Fig. 18), three sets of signals at 8.46, 8.43, 8.41, 8.06, 7.81 and 7.49 ppm were observed in the ¹H NMR spectrum, and have been assigned to the protons within the pyridyl groups. Additionally, the signals at 6.25, 6.14, 6.05, 5.92, 3.04, 2.95, 2.38, 2.27, 1.52 and 1.44 ppm were assigned to the p-cymene protons. Similar to compound **2**(OTf)₁₂, in the ¹H NMR spectrum of compound **3**(OTf)₁₂ (Supplementary Fig. 21), the signals at 8.42, 8.41, 8.39, 7.98, 7.79 and 7.47 ppm were assigned to the pyridyl protons of ligand **L1**, and the signals at 6.25, 6.13, 6.05, 5.92, 3.05, 2.95, 2.41, 2.28, 1.53 and 1.45 ppm were attributed to the protons of the p-cymene moiety.

The recorded ¹H-¹H COSY NMR and ¹H DOSY NMR spectra ($D_2 = 2.34 \times 10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$) of **2**(OTf)₁₂ and **3**(OTf)₁₂ ($D_3 = 2.40 \times 10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$) (Supplementary Figs. 19, 20, 22 and 23) unambiguously supported their formation in solution. Moreover, the calculated hydrodynamic radii of **2**(OTf)₁₂ (16.0 Å) and **3**(OTf)₁₂ (15.7 Å) were consistent with the proposed molecular structures of the compounds (Supplementary Figs. 51 and 52).

Self-assembly of Borromean rings **4**(OTf)₁₂ and **5**(OTf)₁₂.

Previous results have shown that building units with relatively short Ru···Ru distances and small conjugated planes can form Linear [3]catenanes **1**(OTf)₁₂, **2**(OTf)₁₂, and **3**(OTf)₁₂ through

intermolecular $\pi \cdots \pi$ stacking interactions between the central moiety of ligand **L1** and the pyridine rings of another ligand **L1**. We assumed that increasing the π -conjugated area of the building unit could significantly increase the $\pi \cdots \pi$ stacking interactions between the building unit and the ligand **L1**, thereby leading to the formation of structurally more robust supramolecular architectures. Note that the ligand **L1** and the building units must be sufficiently large and in the correct proportions to achieve the desired ring interactions. Therefore, a difference in length between ligand **L1** and the building unit that is twice the characteristic $\pi \cdots \pi$ interaction distance necessitates the combination of equimolar amounts of building units **E4**(OTf)₂ and **E5**(OTf)₂ with 0.12 mmol of **L1**. Methanolic mixtures of **E4**(OTf)₂ and **E5**(OTf)₂ with **L1** were stirred at room temperature, resulting in homogeneous solutions with varying colouration (**4**(OTf)₁₂: 90% and **5**(OTf)₁₂: 87%, Fig. 3a).

Furthermore, a detailed comparison of the spatial configurations between **E4**(OTf)₂ and **E5**(OTf)₂, as well as **E1**(OTf)₂, **E2**(OTf)₂ and **E3**(OTf)₂, revealed a discernible increase in metal-metal separation distances, which occurred concurrently with the extension of the conjugated plane. These structural modifications offered critical insights and a solid foundation for the rational design of more stable supramolecular assemblies in future studies.

The synthesis of two Borromean rings was successfully achieved through multidimensional control of the constructed building blocks, with the resulting compounds being comprehensively characterized by single-crystal X-ray diffraction analysis. The molecular structures of **4**(OTf)₁₂ and **5**(OTf)₁₂ belonged to the *P2/n* and *Fd $\bar{3}$ c* space groups, respectively (Supplementary Tables 4 and 5). Specifically, compound **4**(OTf)₁₂ was formed by three mutually dependent rings, and the absence of any of them resulted in the separation of the other two (Fig. 3b, c). In both structures, three rectangular metallamacrocycles are mechanically interlocked to form a highly symmetrical topology. The metallamacrocycle in **4**(OTf)₁₂ has dimensions of 16.99 $\times$ 8.29 Å, while that in **5**(OTf)₁₂ measures 17.06 $\times$ 8.28 Å (Supplementary Figs. 5d, 5e, 6d, 6e, 7d and 7e). Furthermore, the recorded distances between the building units and the ligand **L1** in compound **4**(OTf)₁₂ ranged from 3.40 Å to 3.54 Å (Fig. 3d). The spatial arrangement of molecular building units was governed by synergistic stacking interactions and the spatial constraints of the building units, resulting in the formation of a stable topological structure (Fig. 3e). Notably, as the conjugated plane of the

building units expanded, no subsequent structural alterations were observed. Furthermore, the stability between each ring was maintained through their inherent interactions. Similarly, the internal $\pi \cdots \pi$ stacking distances of compound **5**(OTf)₁₂ ranged from 3.52 Å to 3.56 Å. (Supplementary Fig. 4).

While the conjugated plane of the building unit increased in size for both raw **E4**(OTf)₂ and **E5**(OTf)₂, no significant structural alterations were observed. Moreover, the presence of intramolecular $\pi \cdots \pi$ stacking interactions among the internal ligands and building units drove the formation of a highly symmetrical dodecanuclear topological architecture in both compounds (Fig. 4a). The behaviour of compounds **4**(OTf)₁₂ and **5**(OTf)₁₂ in solution was studied through ¹H NMR, ¹H-¹H COSY NMR and ¹H DOSY NMR spectroscopy. In the ¹H NMR spectrum of compound **4**(OTf)₁₂ (Supplementary Fig. 24), two doublets were observed at 9.17 and 8.08 ppm, which were attributed to the protons of the pyridyl groups of ligand **L1**. Furthermore, the resonances corresponding to the protons of the naphthyl moiety appeared at 6.67 ppm, the phenyl protons of the p-cymene group appeared at 5.90 and 5.74 ppm, and the alkyl protons of the p-cymene group appeared at 2.86, 2.34 and 1.32 ppm, respectively. Moreover, the recorded diffusion coefficient ($D_4 = 2.64 \times 10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$) and ¹H-¹H COSY NMR spectrum (Supplementary Figs. 25 and 26) of **4**(OTf)₁₂ were consistent with the proposed structure. The ¹H NMR spectrum of compound **5**(OTf)₁₂ was very similar to that of **4**(OTf)₁₂ and exhibited signals corresponding to the protons of the pyridyl group at 9.01 and 8.55 ppm, as well as resonances at 7.67, 7.44, 6.03, 5.96, 2.55, 2.17 and 1.45 ppm belonging to the protons of building unit **E5**(OTf)₂ (Supplementary Fig. 27). Notably, the hydrodynamic radii of all the reported Borromean rings (**4**(OTf)₁₂: 14.2 Å and **5**(OTf)₁₂: 12.0 Å) were determined by using the diffusion coefficients from ¹H DOSY NMR spectra ($D_5 = 3.14 \times 10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$) (Fig. 4b, c and Supplementary Fig. 29) and were in good agreement with the SC-XRD data (Supplementary Figs. 53 and 54). Additionally, the ESI-TOF-MS analysis revealed strong peaks at $m/z = 3460.39$ ($[\mathbf{1}(\text{OTf})_{12} - 2\text{OTf}]^{2+}$) (Fig. 4d), $m/z = 1354.81$ ($[\mathbf{4}(\text{OTf})_{12} - 5\text{OTf}]^{5+}$) (Fig. 4e) and $m/z = 1891.30$ ($[\mathbf{5}(\text{OTf})_{12} - 4\text{OTf}]^{4+}$) (Supplementary Fig. 57) for the respective compounds. NMR analysis revealed the behaviour of each compound in solution.

Conversely, ESI-TOF-MS analysis confirmed that it was consistent with the dodecanuclear supramolecular structure disclosed by SC-XRD.

Next, the effects of concentration variations on the structural integrity of Borromean rings **4**(OTf)₁₂ and **5**(OTf)₁₂ were examined. Remarkably, no significant changes were detected in the ¹H NMR spectra of **4**(OTf)₁₂ and **5**(OTf)₁₂ when their concentrations were increased or decreased in CD₃OD (Supplementary Figs. 33 - 36). These outcomes indicated that compounds **4**(OTf)₁₂ and **5**(OTf)₁₂ retain high stability in solution without experiencing structural changes that are dependent on concentration.

Self-assembly of Solomon link **6**(OTf)₈.

We have demonstrated that Linear [3]catenanes and Borromean rings can be synthesized using building units of varying sizes and π -conjugation planes while employing the same ligand precursor **L1**. Synthesizing a novel Solomon link structure using ligand **L1** might present greater challenges for the design of the building units. The formation of Solomon links typically involves the mutual twisting of interlocked metallamacrocycles, which are stabilized by $\pi \cdots \pi$ stacking interactions. However, the intrinsic rigidity of the ligand precursor **L1** could impede torsional adjustments between its constituent rings. This topological feature of **L1** prompted us to focus our attention on the use of hinge-containing building units with controlled rotational freedom, achievable through silver ion coordination known to stabilize topological conformations and provide rotational flexibility,^{48,49} in an attempt to obtain a molecular Solomon link through dynamic conformational modulation (Fig. 5a). The building unit **E6**(OTf)₂ was synthesized using 2-(1H-benzimidazol-2-yl)-1H-benzimidazole (BiBzIm), which possessed multiple coordination sites, and was subsequently mixed with the ligand precursor **L1** in methanol. Yellow crystals of compound **6**(OTf)₈, suitable for X-ray diffraction analysis, were obtained by diffusing isopropyl ether into a methanolic mixture of **E6**(OTf)₂ and ligand **L1** (0.12 mmol) at room temperature (Supplementary Table 6). Each metallamacrocycle experienced structural twisting, which was then followed by interlocking to create a heterometallic Solomon link topology (Fig. 5b, c, d). Each macrocyclic component has dimensions of 17.03 $\times$ 11.74 Å (Supplementary Figs. 5f, 6f and 7f). Furthermore, the measured distances between the building unit **E6**(OTf)₂ and the ligand **L1** ranged from 3.19 Å to 3.37 Å within the molecular structure of compound **6**(OTf)₈ (Fig. 5e). Additionally, the lateral and longitudinal dimensions of **6**(OTf)₈ were 22.03 Å and 12.58 Å, respectively (Fig. 5f). Moreover, multiple $\pi \cdots \pi$ stacking interactions cooperatively stabilized its octanuclear

topological architecture.

The structure of compound **6**(OTf)₈ was also characterized by ¹H NMR (Supplementary Fig. 30), ¹H-¹H COSY NMR (Supplementary Fig. 31), and ¹H DOSY NMR (Supplementary Fig. 32) spectroscopy. Distinct sets of resonances appeared in the ¹H NMR spectrum of **6**(OTf)₈ at 7.12, 7.00, 6.78, and 6.39 ppm and were assigned to the protons of the pyridine groups. The signals corresponding to the protons of **E6**(OTf)₂ appeared at 8.44, 8.33, 8.17, 7.94, 7.85, 7.66, 7.58, and 7.43 ppm, while the resonances of the p-cymene protons were recorded at 7.72, 6.69, 6.23, 5.41, 2.26, 2.16, 1.87, 1.73, 1.29, and 1.17 ppm. Interestingly, the dimensions of **6**(OTf)₈, as determined by SC-XRD data (Supplementary Fig. 55), were in close agreement with the hydrodynamic radii (13.8 Å) calculated from ¹H DOSY NMR spectroscopy ($D_6 = 2.72 \times 10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$). Moreover, the observed signal at $m/z = 1247.18$ in the ESI-TOF-MS spectrum of **E6**(OTf)₂ was attributed to the [**6**(OTf)₈-5OTf]⁵⁺ ion (Supplementary Fig. 58), which further supported the crystallographic data.

Photothermal conversion studies.

In recent years, materials featuring outstanding photothermal properties have found remarkable applications in photothermal therapy, solar light-induced seawater desalination, photothermal actuation or dynamic camouflage, among other fields.^{50,51} Notably, supramolecular compounds with multiple $\pi \cdots \pi$ stacking interactions significantly enhance photothermal conversion efficiency, reinforce topological stability, advance biological applications, and enable efficient energy transduction.^{52,53} Subsequently, the photothermal conversion properties of ligand precursor **L1**, building units **E1**(OTf)₂ - **E6**(OTf)₂, and compounds **1**(OTf)₁₂ - **6**(OTf)₈ were systematically investigated (Supplementary Fig. 65). To compare equivalent stacking quantities and achieve a 2:2:2:2:2:3 molar ratio of the compounds, their absorption profiles were systematically recorded across relevant wavelengths, yielding comprehensive UV-Vis spectral data. Notably, compound **4**(OTf)₁₂ exhibited the strongest absorption intensity (Fig. 6a). We evaluated the photothermal conversion performance of these compounds under 730 nm laser irradiation (Fig. 6b). Comparative photothermal analysis revealed distinct performance variations: Borromean rings **4**(OTf)₁₂ and **5**(OTf)₁₂ exhibited superior photothermal response compared to Linear [3]catenanes **1**(OTf)₁₂, **2**(OTf)₁₂ and **3**(OTf)₁₂, while Solomon link **6**(OTf)₈ demonstrated minimal activity (Supplementary Figs. 66 - 71). Remarkably, compound **4**(OTf)₁₂ exhibited the most substantial

temperature increase ($\Delta T = 132.8\text{ }^{\circ}\text{C}$) upon irradiation. To further substantiate the photothermal conversion results with experimental data, the photothermal performance of the compounds in solution was systematically evaluated, and the corresponding conversion efficiency was calculated on the basis of the UV-Vis absorption properties of the solutions (**1**(OTf)₁₂: 0.368; **2**(OTf)₁₂: 0.396; **3**(OTf)₁₂: 0.444; **4**(OTf)₁₂: 0.652; **5**(OTf)₁₂: 0.490; **6**(OTf)₈: 0.235) (Fig. 6c). Interestingly, compound **4**(OTf)₁₂ showed the most sensitive photothermal response (Fig. 6d and Supplementary Figs. 72 - 77) and compound **4**(OTf)₁₂ showed approximately the same heating trend during ten cycles of testing (Fig. 6e, f). Moreover, compounds **1**(OTf)₁₂ - **6**(OTf)₈ exhibited conversion efficiencies of 35% - 43%, 52% - 54%, 54% - 61%, 77% - 81%, 65% - 73%, and 23% - 30%, respectively (Supplementary Figs. 78 - 83). Note that thermogravimetric analysis revealed the frameworks of compounds **1**(OTf)₁₂, **2**(OTf)₁₂, **3**(OTf)₁₂, **4**(OTf)₁₂, **5**(OTf)₁₂, and **6**(OTf)₈ began to collapse at 214 $^{\circ}\text{C}$, 205 $^{\circ}\text{C}$, 236 $^{\circ}\text{C}$, 191 $^{\circ}\text{C}$, 208 $^{\circ}\text{C}$, and 304 $^{\circ}\text{C}$, respectively, demonstrating their good thermal stability (Supplementary Figs. 59 - 64). All compounds retained outstanding photothermal performance after 10 cycles, mirroring the trend initially reported for the highest-performing sample (Supplementary Fig. 84), compared with other reported materials, our supramolecular architectures demonstrate a promising photothermal conversion efficiency (Supplementary Table 7), underscoring their significant potential.

Comparative EPR spectroscopic measurements were then conducted on the building units and their corresponding compounds. Intense EPR signals indicated the presence of unpaired electrons from the organic radicals of the building units, consistent with the ground-state charge-transfer interactions observed in photothermal conversion studies.^{54,55} EPR spectroscopy signal changes before and after light exposure were observed for each compound. Moreover, the signal intensities of compounds **1**(OTf)₁₂ - **6**(OTf)₈ was increased by 4.59, 5.08, 6.58, 11.66, 6.64, and 3.67 times, respectively (Fig. 7). The results indicated that the organic radical properties of the components directly govern the photoresponsive behaviour of the compounds.

Discussion

In summary, a systematic ligand design approach enabled the preparation of a wide range of supramolecular architectures, demonstrating how specific ligand modifications direct structural topology. This method was based on fine-tuning of the structural features and electronic properties of the building units used. Consequently, three Linear [3]catenanes **1**(OTf)₁₂, **2**(OTf)₁₂, and **3**(OTf)₁₂; two Borromean rings **4**(OTf)₁₂, and **5**(OTf)₁₂; and one heterometallic Solomon link **6**(OTf)₁₂ were rationally obtained in high yields. The molecular structures of all the compounds were unambiguously confirmed through SC-XRD analysis, NMR spectroscopy, and high-resolution ESI-TOF-MS analysis. Remarkably, the compounds displayed exceptional structural stability and photothermal response properties, which could be attributed to the presence of multiple $\pi \cdots \pi$ stacking interactions and the free radical effect of the building units. EPR spectroscopic data provided significant mechanistic insights into their photothermal conversion behaviour. The obtained data provide an important groundwork for the production of next-generation materials that feature both exceptional stability and remarkable photothermal performance.

Methods

Materials. Reagents and solvents were acquired from commercial suppliers and employed as received, unless specifically stated otherwise.

Synthesis of ligand L1. A mixture of dithiooxamide (180.3 mg, 1.5 mmol) and 4-Pyridinecarboxaldehyde (0.20 mL, 2.2 mmol) in anhydrous DMF (20 mL) was refluxed at 150°C for 6 hours. Upon cooling, The product was washed with deionized water. The resulting yellow solid was isolated by filtration, and obtained in 76% yield.

Synthesis of E1(OTf)₂, E2(OTf)₂, E3(OTf)₂, E4(OTf)₂, and E5(OTf)₂. Adding AgOTf (123.2 mg, 0.48 mmol) and Dichloro(p-cymene)ruthenium(II) dimer (73.49 mg, 0.12 mmol) to 18 ml methanol at room temperature, the mixture was treated with light protection and stirred in dark condition for 12 h, and then the white AgCl precipitate was filtered out to obtain yellow solution. Subsequently 2,5-Dihydroxy-1,4-Benzoquinone (16.8 mg, 0.12 mmol), 2,5-Dichloro-3,6-Dihydroxy-1,4-Benzoquinone (25.1 mg, 0.12 mmol), 2,5-Dibromo-3,6-Dihydroxy-1,4-Benzoquinone (35.7 mg, 0.12 mmol), 5,8-Dihydroxy-1,4-naphthoquinone (22.8 mg, 0.12 mmol)

or 6,11-Dihydroxy-5,12-Naphthacenedione (34.8 mg, 0.12 mmol), and NaOH (9.6 mg, 0.24 mmol) were added to the filtrate. Then the mixed solution was stirred at room temperature for 12 h to give different colors solution, the nonpolar solvent ether was added to the mixed solution of methanol, the product was recrystallized from a diethyl ether mixture to afford **E1(OTf)₂**, **E2(OTf)₂**, **E3(OTf)₂**, **E4(OTf)₂**, and **E5(OTf)₂**.

Synthesis of E6(OTf)₂. A mixture of AgOTf (184.8 mg, 0.72 mmol) and dichloro(p-cymene)ruthenium(II) dimer (73.49 mg, 0.12 mmol) in methanol (18 mL) was stirred under light protection at room temperature for 12 h. Subsequently, the white AgCl precipitate was removed by filtration to afford a yellow solution. This filtrate was then charged with 2-(1H-benzimidazol-2-yl)-1H-benzimidazole (28.1 mg, 0.12 mmol) and NaOH (9.6 mg, 0.24 mmol). Then the mixed solution was stirred at room temperature for 12 h to give a yellow solution, the product was precipitated by adding diethyl ether to the methanol solution, collected by filtration, and then recrystallized from diethyl ether to afford **E6(OTf)₂**.

Synthesis of 1(OTf)₁₂, 2(OTf)₁₂, 3(OTf)₁₂, 4(OTf)₁₂, 5(OTf)₁₂, and 6(OTf)₈. Mixing the ligand **L1** with compounds **E1(OTf)₂**, **E2(OTf)₂**, **E3(OTf)₂**, **E4(OTf)₂**, **E5(OTf)₂** or **E6(OTf)₂** in equal molar amount in 18ml methanol solution for 12h. The single crystals were grown by liquid diffusion, and the nonpolar solvent ether was added to the mixed solution of methanol. The product was recrystallized from a diethyl ether mixture to afford crystals **1(OTf)₁₂**, **2(OTf)₁₂**, **3(OTf)₁₂**, **4(OTf)₁₂**, **5(OTf)₁₂**, and **6(OTf)₈**. Notably, single crystals of compound **6(OTf)₈** suitable for X-ray diffraction were obtained via diffusion after just one day, whereas compounds **1(OTf)₁₂**, **2(OTf)₁₂**, and **3(OTf)₁₂** required two or three days, and compounds **4(OTf)₁₂** and **5(OTf)₁₂** required five to seven days.

Characterization data for 1(OTf)₁₂ (Linear [3]catenane). Total yield of crystals: 89%, ¹H NMR (500 MHz, CD₃OD, ppm, with respect to p-cymene Ru): δ = 8.44 (d, 2H J = 7.5 Hz, pyridyl-H_{a1}), δ = 8.41 (d, 2H J = 9.0 Hz, pyridyl-H_{a2}), δ = 8.38 (d, 2H J = 7.0 Hz, pyridyl-H_{a3}), δ = 8.03 (d, 2H J = 7.5 Hz, pyridyl-H_{b1}), δ = 7.72 (d, 2H J = 7.5 Hz, pyridyl-H_{b3}), δ = 7.39 (d, 2H J = 7.0 Hz, pyridyl-H_{b2}), δ = 6.17 (d, 2H J = 7.5 Hz, phenyl-H_{f1} of p-cymene), δ = 6.05 (d, 2H J = 7.5 Hz, phenyl-H_{f2} of p-cymene), δ = 6.02 (s, 1H, benzoquinonyl-H_{c1}), δ = 5.95 (d, 2H J = 7.5 Hz, phenyl-H_{e1} of p-cymene), δ = 5.83 (d, 2H J = 6.5 Hz, phenyl-H_{e2} of p-cymene), δ = 5.78 (s, 1H, benzoquinonyl-H_{c2}), δ = 3.00 (m, 1H J = 34.5 Hz, substituted-methyl-H_{g1} of p-cymene), δ = 2.90 (m, 1H J = 33.5 Hz, substituted-methyl-H_{g2} of p-cymene), δ = 2.31 (s, 3H, methyl-H_{d1} of p-cymene), δ = 2.20 (d, 3H J =

12.0 Hz, methyl- H_{d2} of p-cymene), $\delta = 1.46$ (d, 6H $J = 8.5$ Hz, methyl- H_{h1} of p-cymene), $\delta = 1.38$ (d, 6H $J = 8.5$ Hz, methyl- H_{h2} of p-cymene); ESI-TOF-MS (m/z): $[1(OTf)_{12}-2OTf]^{2+}$ calcd. for $C_{252}H_{228}O_{60}N_{24}S_{24}F_{36}Ru_{12}$, 3460.3936; found, 3460.3938; analysis (calcd., found for $C_{252}H_{228}O_{60}N_{24}S_{24}F_{36}Ru_{12}$): C (41.93, 41.64), H (3.18, 3.23), N (4.66, 4.78).

Characterization data for 2(OTf)₁₂ (Linear [3]catenane). Total yield of crystals: 90%. 1H NMR (500 MHz, CD_3OD , ppm, with respect to p-cymene Ru): $\delta = 8.46$ (d, 2H $J = 6.5$ Hz, pyridyl- H_{a1}), $\delta = 8.43$ (d, 2H $J = 6.0$ Hz, pyridyl- H_{a2}), $\delta = 8.41$ (d, 2H $J = 6.5$ Hz, pyridyl- H_{a3}), $\delta = 8.06$ (d, 2H $J = 6.5$ Hz, pyridyl- H_{b1}), $\delta = 7.81$ (d, 2H $J = 6.5$ Hz, pyridyl- H_{b3}), $\delta = 7.49$ (d, 2H $J = 5.5$ Hz, pyridyl- H_{b2}), $\delta = 6.25$ (d, 2H $J = 6.0$ Hz, phenyl- H_{e1} of p-cymene), $\delta = 6.14$ (d, 2H $J = 6.0$ Hz, phenyl- H_{e2} of p-cymene), $\delta = 6.05$ (d, 2H $J = 6.0$ Hz, phenyl- H_{d1} of p-cymene), $\delta = 5.92$ (d, 2H $J = 6.0$ Hz, phenyl- H_{d2} of p-cymene), $\delta = 3.04$ (m, 1H $J = 27.5$ Hz, substituted-methyl- H_{f1} of p-cymene), $\delta = 2.95$ (m, 1H $J = 26.5$ Hz, substituted-methyl- H_{f2} of p-cymene), $\delta = 2.38$ (s, 3H, methyl- H_{c1} of p-cymene), $\delta = 2.27$ (s, 3H, methyl- H_{c2} of p-cymene), $\delta = 1.52$ (d, 6H $J = 7.0$ Hz, methyl- H_{g1} of p-cymene), $\delta = 1.44$ (d, 6H $J = 6.5$ Hz, methyl- H_{g2} of p-cymene); analysis (calcd., found for $C_{252}H_{216}O_{60}N_{24}S_{24}F_{36}Cl_{12}Ru_{12}$): C (39.66, 39.95), H (2.85, 2.73), N (4.40, 4.26).

Characterization data for 3(OTf)₁₂ (Linear [3]catenane). Total yield of crystals: 91%. 1H NMR (500 MHz, CD_3OD , ppm, with respect to p-cymene Ru): $\delta = 8.42$ (d, 2H $J = 2.0$ Hz, pyridyl- H_{a1}), $\delta = 8.41$ (d, 2H $J = 7.0$ Hz, pyridyl- H_{a2}), $\delta = 8.39$ (d, 2H $J = 2.0$ Hz, pyridyl- H_{a3}), $\delta = 7.98$ (d, 2H $J = 4.0$ Hz, pyridyl- H_{b1}), $\delta = 7.79$ (d, 2H $J = 7.0$ Hz, pyridyl- H_{b3}), $\delta = 7.47$ (d, 2H $J = 5.0$ Hz, pyridyl- H_{b2}), $\delta = 6.25$ (d, 2H $J = 6.0$ Hz, phenyl- H_{e1} of p-cymene), $\delta = 6.13$ (d, 2H $J = 6.0$ Hz, phenyl- H_{e2} of p-cymene), $\delta = 6.05$ (d, 2H $J = 6.0$ Hz, phenyl- H_{d1} of p-cymene), $\delta = 5.92$ (d, 2H $J = 6.0$ Hz, phenyl- H_{d2} of p-cymene), $\delta = 3.05$ (m, 1H $J = 14.0$ Hz, substituted-methyl- H_{f2} of p-cymene), $\delta = 2.95$ (m, 1H $J = 28.5$ Hz, substituted-methyl- H_{f1} of p-cymene), $\delta = 2.41$ (s, 3H, methyl- H_{c1} of p-cymene), $\delta = 2.28$ (s, 3H, methyl- H_{c2} of p-cymene), $\delta = 1.53$ (d, 6H $J = 7.0$ Hz, methyl- H_{g1} of p-cymene), $\delta = 1.45$ (d, 6H $J = 7.0$ Hz, methyl- H_{g2} of p-cymene); analysis (calcd., found for $C_{252}H_{216}O_{60}N_{24}S_{24}F_{36}Br_{12}Ru_{12}$): C (37.07, 37.36), H (2.67, 2.42), N (4.12, 3.96).

Characterization data for 4(OTf)₁₂ (Borromean ring). Total yield of crystals: 90%. 1H NMR (500 MHz, CD_3OD , ppm, with respect to p-cymene Ru): $\delta = 9.16$ (d, 2H $J = 5.0$ Hz, pyridyl- H_a), $\delta = 8.08$ (d, 2H $J = 5.0$ Hz, pyridyl- H_b), $\delta = 6.67$ (s, 2H, naphthyl- H_c), $\delta = 5.90$ (d, 2H $J = 5.5$ Hz, phenyl- H_f of p-cymene), $\delta = 5.74$ (d, 2H

$J = 5.5$ Hz, phenyl- H_e of p-cymene), $\delta = 2.86$ (s, 1H, substituted-methyl- H_g of p-cymene), $\delta = 2.24$ (s, 3H, methyl- H_d of p-cymene), $\delta = 1.32$ (s, 6H, methyl- H_h of p-cymene); ESI-TOF-MS (m/z): $[4(OTf)_{12}-5OTf]^5+$ calcd. for $C_{276}H_{240}O_{60}N_{24}S_{12}F_{36}Ru_{12}$, 1354.8094; found, 1354.8053 analysis (calcd., found for $C_{276}H_{240}O_{60}N_{24}S_{12}F_{36}Ru_{12}$): C (46.46, 46.13), H (3.39, 3.48), N (4.71, 4.63).

Characterization data for 5(OTf)₁₂ (Borromean ring). Total yield of crystals: 87%. 1H NMR (500 MHz, CD_3OD , ppm, with respect to p-cymene Ru): $\delta = 9.01$ (d, 2H $J = 2.0$ Hz, pyridyl- H_a), $\delta = 8.55$ (d, 2H $J = 6.5$ Hz, tetracene- H_c , $\delta = 7.67$ (d, 2H $J = 7.5$ Hz, tetracene- H_d), $\delta = 7.44$ (d, 2H $J = 5.5$ Hz, pyridyl- H_b), $\delta = 6.03$ (d, 2H $J = 5.0$ Hz, phenyl- H_g of p-cymene), $\delta = 5.95$ (d, 2H $J = 9.0$ Hz, phenyl- H_f of p-cymene), $\delta = 2.55$ (s, 3H, methyl- H_e of p-cymene), $\delta = 2.17$ (s, 1H, substituted-methyl- H_h of p-cymene), $\delta = 1.47$ (d, 6H $J = 6.5$ Hz, methyl- H_i of p-cymene); ESI-TOF-MS (m/z): $[5(OTf)_{12}-4OTf]^4+$ calcd. for $C_{324}H_{264}O_{60}N_{24}S_{24}F_{36}Ru_{12}$, 1881.0448; found, 1881.0420 analysis (calcd., found for $C_{324}H_{264}O_{60}N_{24}S_{24}F_{36}Ru_{12}$): C (47.93, 47.73), H (3.28, 3.35), N (4.14, 4.25).

Characterization data for 6(OTf)₈ (Solomon link). Total yield of crystals: 86%. 1H NMR (500 MHz, CD_3OD , ppm, with respect to p-cymene Ru): $\delta = 8.46$ (m, 2H $J = 15.0$ Hz, benzimidazole- H_{c1}), $\delta = 8.33$ (d, 2H $J = 7.5$ Hz, benzimidazole- H_{f1}), $\delta = 8.17$ (d, 2H $J = 8.5$ Hz, benzimidazole- H_{c2}), $\delta = 7.94$ (m, 2H $J = 15.0$ Hz, benzimidazole- H_{f2}), $\delta = 7.85$ (d, 2H $J = 8.5$ Hz, benzimidazole- H_{d1}), $\delta = 7.72$ (d, 2H $J = 5.0$ Hz, phenyl- H_{i2} of p-cymene), $\delta = 7.68$ (d, 2H $J = 7.0$ Hz, benzimidazole- H_{e1}), $\delta = 7.59$ (m, 2H $J = 15.5$ Hz, benzimidazole- H_{d2}), $\delta = 7.43$ (d, 2H $J = 6.5$ Hz, benzimidazole- H_{e2}), $\delta = 7.13$ (m, 2H $J = 15.0$ Hz, pyridyl- H_{a1}), $\delta = 7.01$ (d, 2H $J = 5.0$ Hz, pyridyl- H_{a2}), $\delta = 6.78$ (d, 2H $J = 7.5$ Hz, pyridyl- H_{b1}), $\delta = 6.69$ (d, 2H $J = 6.0$ Hz, phenyl- H_{i1} of p-cymene), $\delta = 6.40$ (d, 2H $J = 5.5$ Hz, pyridyl- H_{b2}), $\delta = 6.23$ (d, 2H $J = 5.5$ Hz, phenyl- H_{h1} of p-cymene), $\delta = 5.42$ (d, 2H $J = 3.5$ Hz, phenyl- H_{h2} of p-cymene), $\delta = 2.67$ (s, 3H, methyl- H_{g1} of p-cymene), $\delta = 2.16$ (s, 6H, methyl- H_{k1} of p-cymene), $\delta = 1.87$ (s, 1H, substituted methyl- H_{j1} of p-cymene), $\delta = 1.73$ (s, 1H, substituted methyl- H_{j2} of p-cymene), $\delta = 1.29$ (s, 3H, methyl- H_{g2} of p-cymene), $\delta = 1.19$ (m, 6H $J = 14.0$ Hz, methyl- H_{k2} of p-cymene); ESI-TOF-MS (m/z): $[6(OTf)_8-5OTf]^5+$ calcd. for $C_{256}H_{208}O_{24}N_{48}S_{16}F_{24}Ag_8Ru_8$, 1246.9790; found, 1246.9770; analysis (calcd., found for $C_{256}H_{208}O_{24}N_{48}S_{16}F_{24}Ag_8Ru_8$): C (44.04, 43.86), H (3.00, 3.13), N (9.63, 9.97).

Data Availability

The X-ray crystallographic data reported in this article has been deposited at the Cambridge Crystallographic

Data Centre (CCDC), under deposition number CCDC 2447354 ($1(\text{OTf})_{12}$), 2503127 ($2(\text{OTf})_{12}$), 2447355 ($3(\text{OTf})_{12}$), 2447356 ($4(\text{OTf})_{12}$), 2447358 ($5(\text{OTf})_{12}$), and 2447360 ($6(\text{OTf})_8$). These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via [www.ccdc.cam.ac.uk/data_request/cif]. The authors declare that all other data supporting the findings of this study are available within the paper and its supplementary information files. All data are available from the corresponding author upon request.

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Author Contributions Statement

J.-X. Y. and X.-Q. W. carried out the synthesis and characterization studies. Ming-Yu Lu and Meng-Yao Lu analyzed the data. M.-Y. Y. and F. A. solved the crystal structure. J.-X. Y. and X.-Q. W. performed photothermal conversion studies. Y. Z., B. L., L.-L. D., and L.-F. M. directed the research. All of the authors contributed to

the analysis of the results and the writing of the manuscript, with grant numbers [22471113] awarded to J.-X. Y. and X.-Q. W., [22171123] to Ming-Yu Lu and Meng-Yao Lu, [242300421139] to M.-Y. Y., F. A. and B. L., [25HASTIT001] to L.-L. D., and [26HASTIT036] Y. Z., and L.-F. M.

Competing Interests Statement

The authors declare no competing interests.

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Figure Legends

Synthesis of compounds $1(\text{OTf})_{12}$, $2(\text{OTf})_{12}$, $3(\text{OTf})_{12}$, $4(\text{OTf})_{12}$, $5(\text{OTf})_{12}$, and $6(\text{OTf})_8$

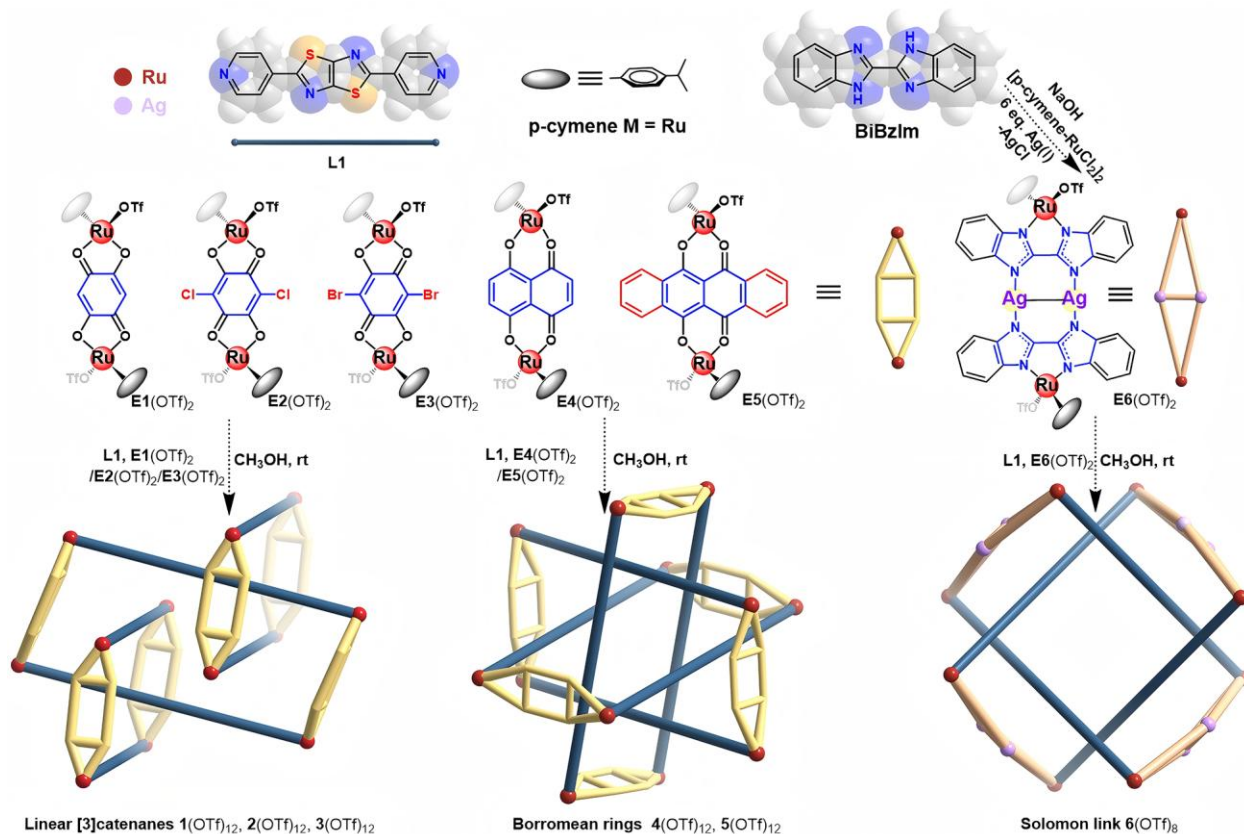


Fig. 1 | Synthesis of Linear [3]catenanes $1(\text{OTf})_{12}$, $2(\text{OTf})_{12}$, and $3(\text{OTf})_{12}$, Borromean rings $4(\text{OTf})_{12}$ and $5(\text{OTf})_{12}$, and Solomon link $6(\text{OTf})_8$.

The crystal structure of Linear [3]catenanes

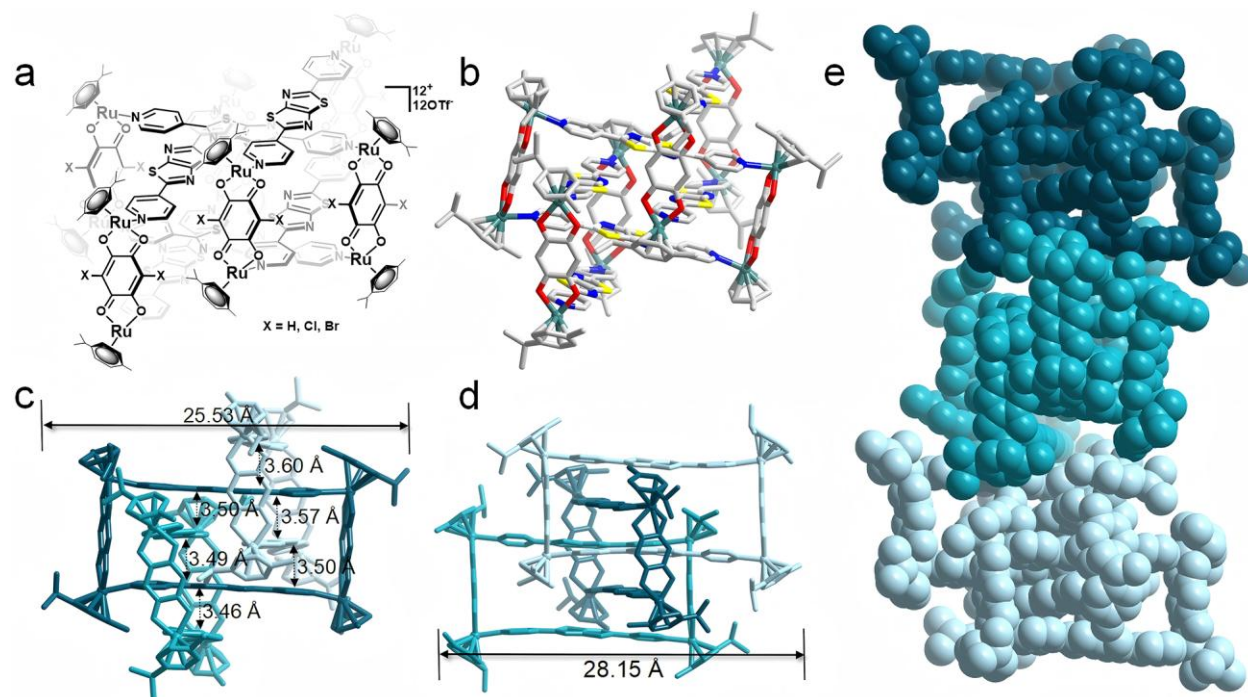


Fig. 2 | (a) Chemical structures of Linear [3]catenanes **1**(OTf)₁₂, **2**(OTf)₁₂, and **3**(OTf)₁₂; (b) Molecular structure of Linear [3]catenanes **1**(OTf)₁₂; (c) Front view, (d) side view, and (e) space-filling representations of Linear [3]catenane **1**(OTf)₁₂ showing $\pi \cdots \pi$ stacking interactions and the front and side distances. Most hydrogen atoms, anions and solvent molecules are omitted for clarity (N, blue; O, red; C, grey; S, yellow; Ru, aqua).

The crystal structure of Borromean rings

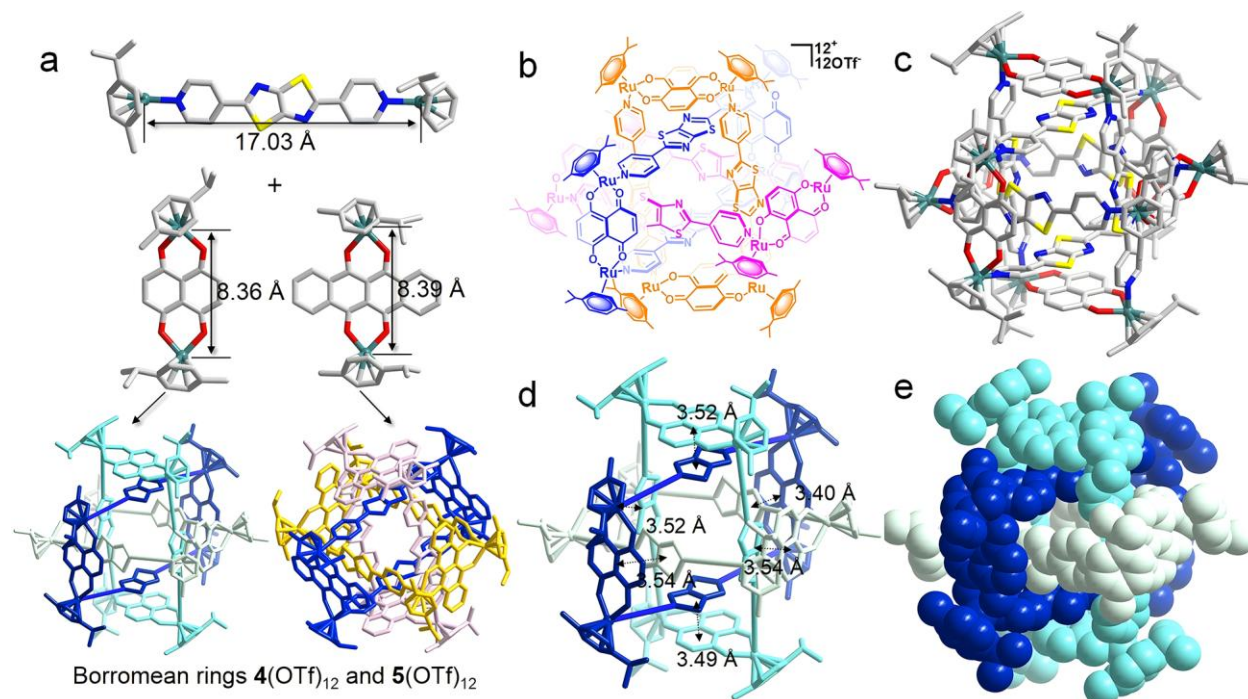


Fig. 3 | (a) Size relationships between ligand **L1** and the building units **E4(OTf)₂** and **E5(OTf)₂** in Borromean rings **4(OTf)₁₂** and **5(OTf)₁₂**; (b) Chemical structure, (c) molecular structure, (d) three colours showing the $\pi \cdots \pi$ stacking interactions and (e) space-filling representations of Borromean ring **3(OTf)₁₂**. Most hydrogen atoms, anions and solvent molecules are omitted for clarity (N, blue; O, red; C, grey; S, yellow; Ru, aqua).

NMR and ESI-TOF-MS experiment of compounds $1(\text{OTf})_{12}$ and $4(\text{OTf})_{12}$

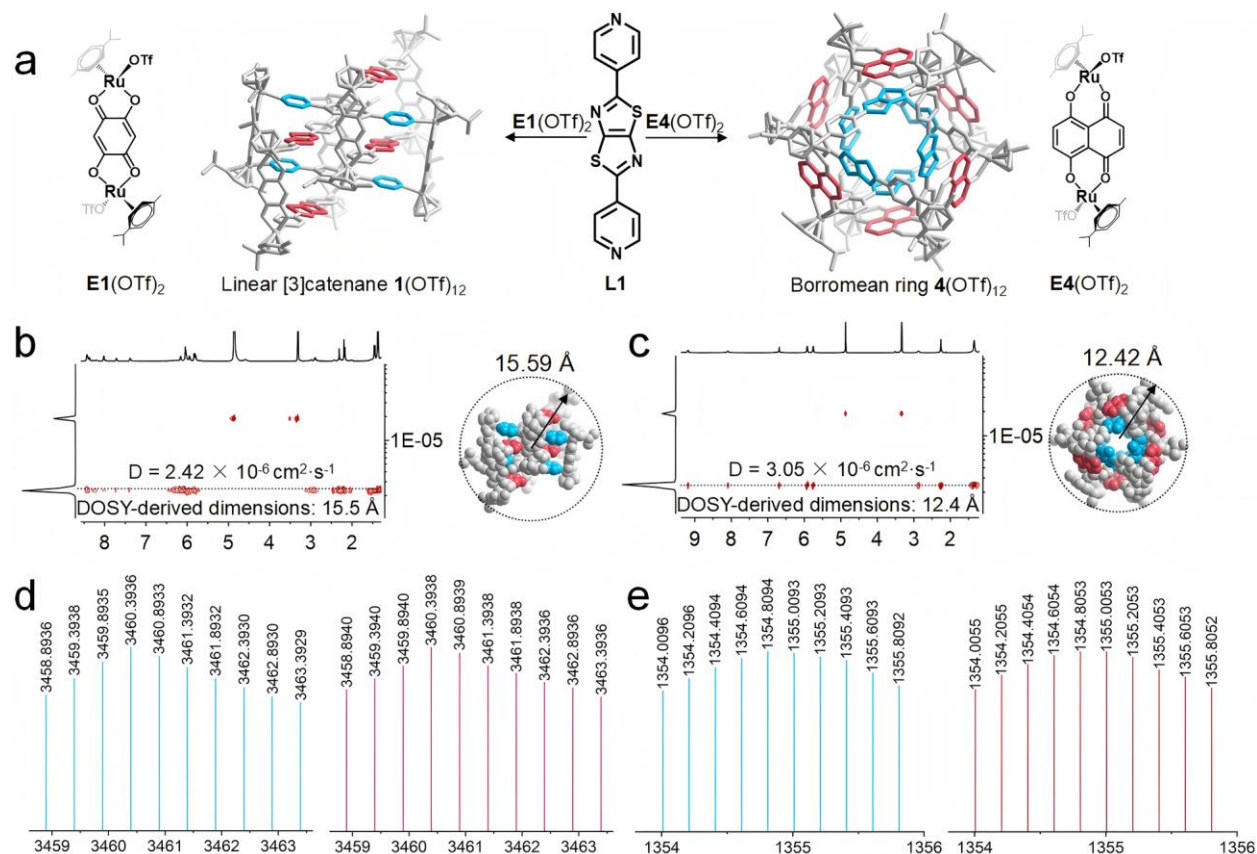


Fig. 4 | (a) Schematic diagram for synthesizing Linear [3]catenane $1(\text{OTf})_{12}$ and Borromean ring $4(\text{OTf})_{12}$; (b) Partial ^1H DOSY NMR spectrum (500 MHz, CD_3OD , 298 K, 15.0 mM with respect to p-cymene Ru, $D_1 = 2.42 \times 10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$) of $1(\text{OTf})_{12}$; (c) Partial ^1H DOSY NMR spectrum (500 MHz, CD_3OD , 298 K, 15.0 mM with respect to p-cymene Ru, $D_4 = 3.05 \times 10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$) of $4(\text{OTf})_{12}$; (d, e) ESI-TOF-MS spectra of compounds $[1(\text{OTf})_{12}-2\text{OTf}]^{2+}$ and $[4(\text{OTf})_{12}-5\text{OTf}]^{5+}$, respectively.

The crystal structure of Solomon link

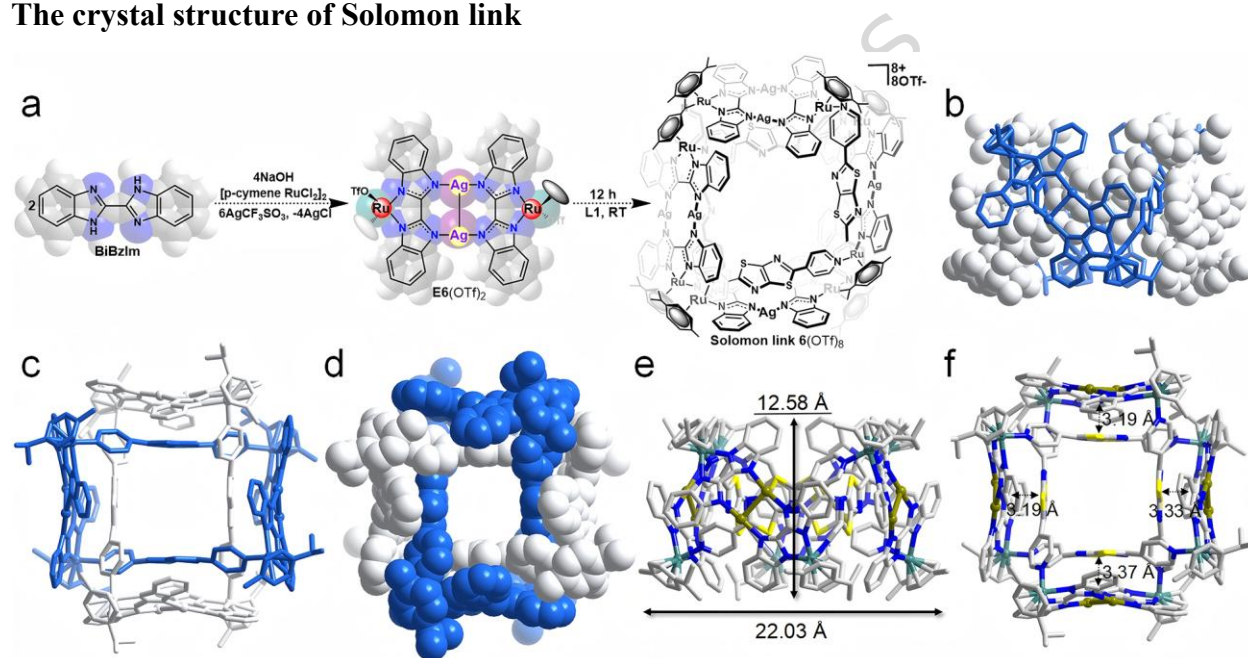


Fig. 5 | (a) Side view, (b) synthesis, (d) top view, (e) two colours and (c, f) space-filling representations of Solomon link $6(\text{OTf})_8$. Most hydrogen atoms, anions and solvent molecules are omitted for clarity (N, blue; O, red; C, grey; S, yellow; Ag, dark yellow; Ru, aqua).

UV-Vis and NIR photothermal conversion experiments

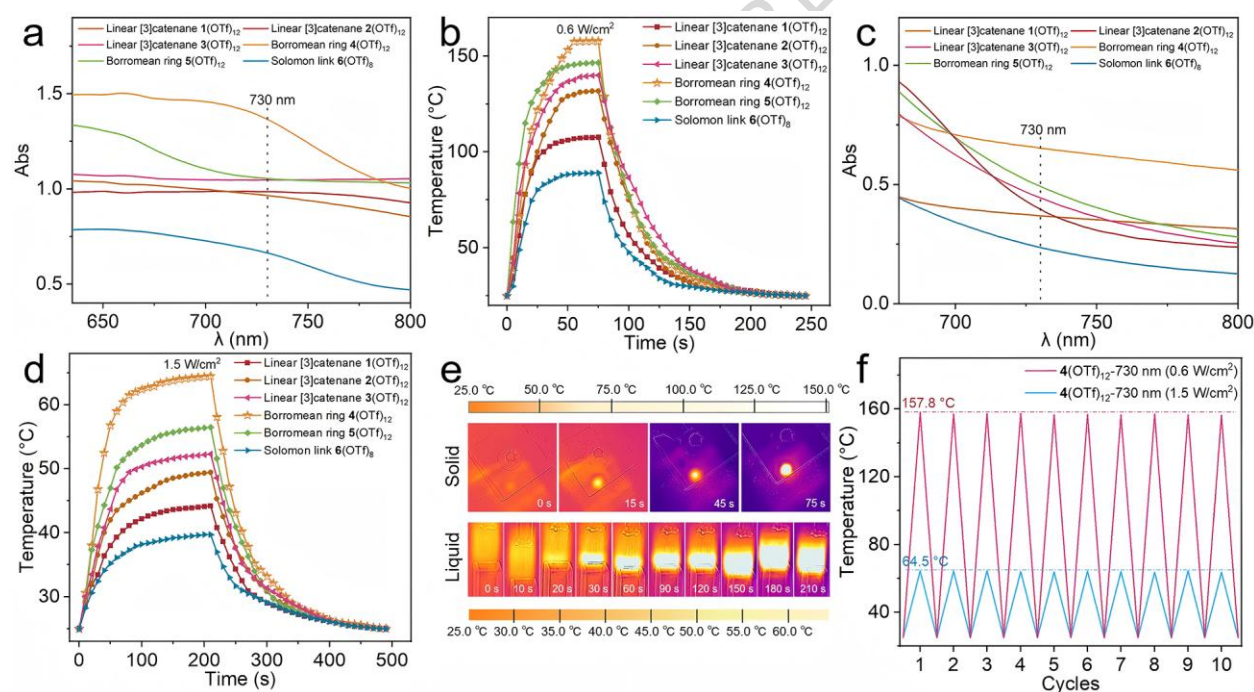


Fig. 6 | (a) UV-Vis absorption of compounds **1(OTf)₁₂** - **6(OTf)₈** at 635 - 800 nm; (b) Photothermal conversion curves of compounds **1(OTf)₁₂** - **6(OTf)₈** in the solid state at 0.6 W/cm²; (c) UV-Vis absorption of compounds **1(OTf)₁₂** - **6(OTf)₈** at 680 - 800 nm in methanol; (d) Photothermal conversion curves of compounds **1(OTf)₁₂** - **6(OTf)₈** at 1.5 W/cm² in methanol; (e) NIR thermal images of compound **4(OTf)₁₂** in solid and liquid states under 730 nm laser irradiation; (f) Ten cycles of heating-cooling of compound **4(OTf)₁₂** upon 730 nm (0.6 W/cm² and 1.5 W/cm²) laser irradiation.

EPR experiment

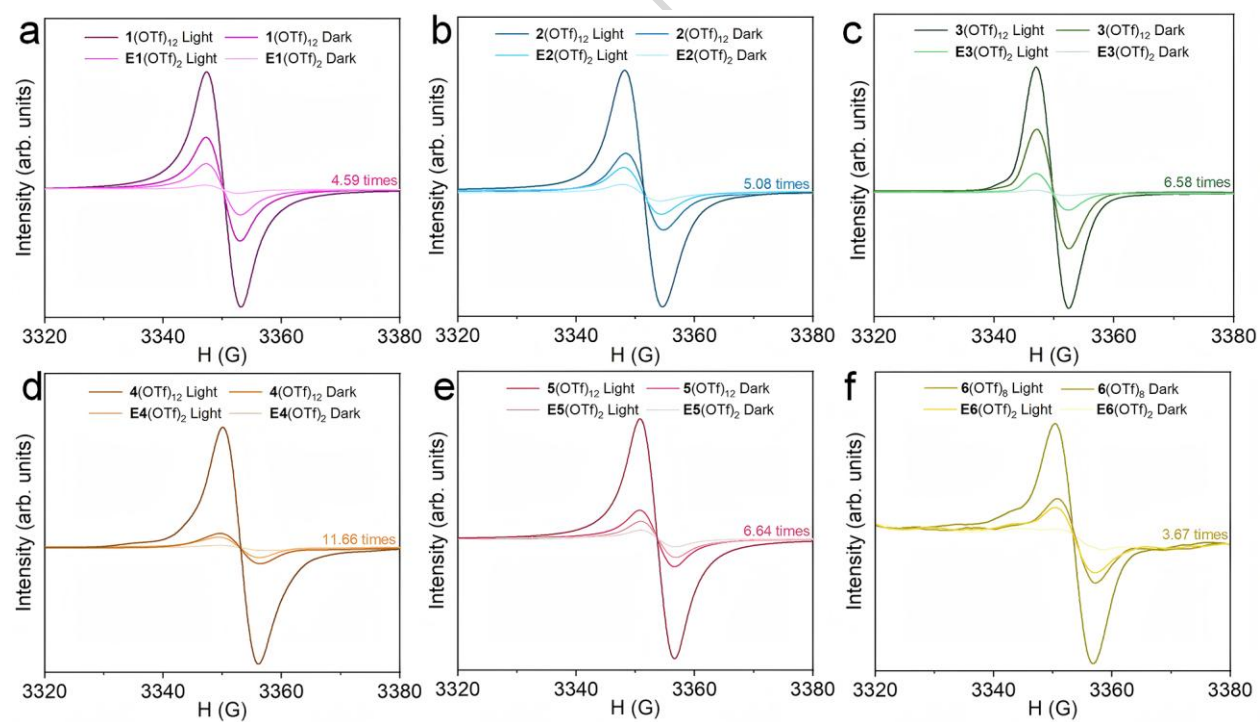


Fig. 7 | The EPR spectra of compounds (a) $1(\text{OTf})_{12}$, (b) $2(\text{OTf})_{12}$, (c) $3(\text{OTf})_{12}$, (d) $4(\text{OTf})_{12}$, (e) $5(\text{OTf})_{12}$ and (f) $6(\text{OTf})_8$ (100 Hz, 298 K).

Editorial Summary:

Mechanically interlocked supramolecular compounds have garnered widespread attention but it remains challenging to investigate topological adjustments in such complicated structures. Here, the authors use a bidentate pyridyl ligand featuring an extended π -conjugated plane for coordination-directed self-assembly to engineer three different kinds of sophisticated complexes.

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